Real corpus analysis: the organizer dataset

The organizer supplied a capture set for the thermal-stand protocol. It is committed under `tests/corpus/real/` and analysed with the same pipeline the application uses, so these numbers describe the shipped files and the shipped code. The generated report is `docs/DATASET_INVESTIGATION.md`; the reproduction command is `python -m scripts.investigate_dataset`.

1. What the dataset contains

File	Role
`research/01_configuration.pcap`	configuration changes and measurements, two TCP sessions
`research/02_variations.pcapng`	other settings, negative temperatures, labels of several lengths, Cyrillic, rejected requests
`research/03_combined_messages.pcap`	several actions in one client send, different packet splits
`additional/04_new_observations.pcapng`	a second experiment with other values and batch sizes
`diagnostics/01..08`	variants of the third record, each with one capture or integrity condition
`action_journal.csv`	109 actions with parameters, time and the result the client showed
`dataset_manifest.json`	the set composition and data origin

The organizer's own `README_RU.md` states the origin: the application bytes come from a real client/simulator TCP socket exchange on localhost, while the Ethernet/IP/TCP framing, segmentation and clock are generated. The set is a protocol-function check, not a passive capture of a device.

`dataset_manifest.json` records the expected session and action counts per research capture. The pipeline reproduces the session counts exactly.

2. Sessions and streams

Capture	Sessions	Stream bytes	Journal actions
`research/01_configuration.pcap`	2	443	14
`research/02_variations.pcapng`	1	608	15
`research/03_combined_messages.pcap`	1	358	8
`additional/04_new_observations.pcapng`	1	384	8
total	5	1793	45

The session counts match `dataset_manifest.json` (2, 1, 1, 1).

3. Message framing and rules

Two rules describe the protocol: `examples/thermostand_request_v2.json` (direction `A_to_B`) and `examples/thermostand_response_v2.json` (direction `B_to_A`). Both use a length-prefixed frame whose two-byte length field counts the four-byte header, carried as `length_bias: 4`.

Direction	Messages	matched	mismatched	incomplete	ambiguous	uncovered
requests	45	45	0	0	0	0
responses	45	45	0	0	0	0

Every message in the research captures is framed and matched, covering 100 % of the bytes in both directions. The `not_applicable` count in a rule report is the other-direction traffic, which is expected.

Per capture

Capture	Request messages	Response messages
`01_configuration`	14	14
`02_variations`	15	15
`03_combined_messages`	8	8
`04_new_observations`	8	8

4. Fields identified

Field	Type	Offset (request / response)	Note
`magic`	uint16	0 / 0	constant `0xa65c` (42588)
`body_length`	uint16	2 / 2	counts the four-byte header
`message_counter`	uint16	4 / 4	increments per exchange
`message_type`	enum	6 / 6	request 16/32/48/126, response 272/288/304/382
`status`	enum	— / 8	0 ok, 1..4 error codes
`enabled`	uint8	8 / 9	hypothesis, set-config and config replies
`period_ms`	uint16	9 / 10	hypothesis
`threshold_raw`	int16	11 / 12	hypothesis
`label`	string	14 / 15	hypothesis, length in a preceding uint8
`requested_count`	uint8	8 / —	read-values request
`measurement_count`	uint8	— / 9	number of measurement records
`device_time_ms`	array uint32	— / 10	hypothesis, stride 7
`temperature_raw`	array int16	— / 14	hypothesis, stride 7; scale 0.01 °C
`alarm`	array uint8	— / 16	hypothesis, stride 7

The enum values were cross-checked against the journal: each decoded `message_type` corresponds to the action the journal records at the same time.

5. Counterexamples and refinement

The first reading, `examples/thermostand_request_v1.json`, assumes the length field counts the payload only. On the configuration capture it produces six counterexamples: the first message is framed four bytes short, so `magic` reads at the wrong offset and `message_counter` falls outside the message.

Rule	matched	mismatched	incomplete
v1 (payload-only length)	0	5	10
v2 (length includes the header)	45	0	0

The fix is the four-byte header bias (`length_bias: 4`), which v2 carries. The counterexamples are the evidence; they are kept in the investigation, not dropped.

6. Diagnostics

Each diagnostic capture is built to exercise one condition. The table is what the reassembler actually reports.

Capture	Condition	Reported
`01_retransmission`	same-bytes retransmission	deduplicated (one retransmission recorded in `B_to_A`, no duplicate bytes)
`02_reordered`	reordered packets	reassembled in sequence order, no gap
`03_missing_segment`	missing segment	`gap`
`04_snaplen_truncated`	truncated snapshot	`gap`
`05_conflicting_overlap`	conflicting overlap	`ambiguity`
`06_missing_beginning`	no connection start	`syn_seen=False`, sequence roles UNKNOWN, stream still built
`07_checksum_offload`	partial pseudo-header checksum	`checksum_offload` when verification is on
`08_bad_checksum`	invalid checksum	`checksum_offload` when verification is on

Checksum verification is off by default because of offloading; the last two rows appear only with `ignore_checksums=False`.

7. Journal correlation

The action journal holds 109 entries. Applying the request rule to the configuration capture and correlating with the journal gives:

```
14 of 14 messages correlated, 95 journal entries unmatched
```

Every decoded request message maps to the operator action it performs, and the decoded parameters agree with the recorded values. The unmatched entries belong to the other captures, which were not part of that correlation run.

8. Comparison with the synthetic corpus

Property	Synthetic corpus	Organizer dataset
Origin	generated end to end	application bytes from a real socket, framing generated
Protocol	`0x01`-led length-prefixed frame	`0xa65c` magic, header-counting length
Research captures	1 primary + 1 held out	3 research + 1 additional
Defect captures	1 combined	8, one condition each
Messages	280 framed	90 (45 + 45)
Rule outcome	v1 gives 80 counterexamples, v2 clears them	v1 gives 6 counterexamples, v2 clears them

Both corpora exercise the same refinement loop: a naive rule, counterexamples tied to bytes, a versioned fix, and a clean verification afterwards. The organizer set adds finer defect coverage (eight single-condition captures) and a second, independent protocol with a different framing convention.

9. Applicability and limits

- The rules are confirmed over 90 messages in four captures, not proven. The organizer states the framing headers are generated, so header-level behaviour seen here should be re-checked against a passive capture.
- `temperature_raw` is a hypothesis with a scale of 0.01 °C; the positiveness of the scale rests on the journal values, not on a byte-level invariant.
- The measurement arrays use a fixed seven-byte stride; a record of a different length would be reported `incomplete`, not silently mis-read.
- Error replies carry a one-byte body, so the configuration fields after `status` are correctly out of the message.

10. How to reproduce

```
python -m scripts.investigate_dataset          # writes docs/DATASET_INVESTIGATION.md
pytest tests/hypothesis/test_real_corpus.py -v # the end-to-end checks
```